



The most distant KBO

How Japanese astronomers using consumer grade telescopes managed to discover the most distant object in the Kuiper Belt. <https://digit.in/may19-69>



The missing KBOs

New Horizons measured no. of small craters on Pluto, allowing scientists to model object density in area. <https://digit.in/may19-70>

there should be. This sudden fall off of expected objects in the Kuiper Belt, is called the Kuiper Cliff.

THE PROPOSED EXPLANATIONS

One of the earliest explanations was that we have an observational bias considering that the larger and nearer objects are easier to discover. In 2004, a dedicated survey of TNOs using data from the Hubble Space Telescope found evidence that the Kuiper Cliff was real, and not a consequence of observational bias.

Another proposed explanation is that a star could have passed the Solar System, very early in the formation of the planets. This star would have had to pass between 80 to 200 AU of the Sun, causing disruptions in the objects at the edge of the Solar System. Such an event would create the Kuiper Cliff as we know it, resulting in a sharp fall off of objects at a distance of 50 AU. However, a passing star would have significantly disrupted the Oort Cloud as well, and the observed characteristics of the Oort Cloud do not agree to the hypothesis that another star brushed the Solar System in its early days. The passing of a star is a low probability event in any case, and it would have to occur at an exact time to explain the observations.

A number of explanations deal with the exact details of what exactly happened during the formation of the Solar System. These are different models to explain the positions of the gas and ice giants, and their migration over time. Jupiter, Saturn, Uranus and Neptune have the biggest influences in the outer solar system, and their orbits have ultimately led to the large scale structures in the Solar System, including the asteroid belt, the Kuiper Belt, the scattered disk and the Oort Cloud. There are many variations of these proposed models. One version suggests a more tightly packed Solar System in the early days of its formation, with as many as five ice giants roughly the size of Neptune. The bigger planets migrated outwards over the course of the formation of the system, and

most of the ice giants were destroyed because of the gravitational influence of the more massive gas giants. This is a model that could excite the objects in the Kuiper Belt. Another model, called the Nice model which best explains the structures in the outer solar system. According to this model, early in the process of the formation of the Solar System, Jupiter and Saturn had a 1:2 resonance lock. This meant that a periodic gravitational influence, with Saturn going around the Sun twice for every time Jupiter did. The combined action of the gas giants would cause the ice giants of Uranus and Neptune to migrate outwards from the Sun, as well severely perturbing the objects in the Kuiper Belt. The Nice model could explain a Kuiper Cliff at a distance of 48 AU, because of the combined gravitational influence of the gas giants from the period during which they were in a resonance lock. However, the model has not yet been reconciled with all the properties of the objects in the Kuiper Belt. Additionally, there are some erratic TNOs that are not explained adequately by the Nice model.

Astronomers using Newtonian mechanics as early as the 1840s, had predicted the existence of a massive planet in the outer solar system, based on perturbations in the orbit of Uranus. Even the discovery of Neptune could not adequately account for the observed perturbations. It was the decades long search for what was known as Planet X, that resulted in the discovery of Pluto. However, Pluto was not big enough to explain the perturbations. The presence of large TNOs such as Eris, the high tilts of Neptune and Uranus, as well as the retrograde orbit of Triton, a moon of Neptune, all suggest that there is another planet lurking in the outer edges of the Solar System.

In 2008 researchers from Kobe University in Japan, Patryk S Lykawka and Tadashi Mukai published a paper that used accretion disk modelling to propose the existence of a previously undiscovered planet beyond the orbit of Pluto, one of the many papers to

do so. This predicted planet would be slightly larger than the size of Mars, about two thirds the size of the Earth. Such a planet, if discovered, would have cleared its orbit of most smaller objects, and could adequately explain the Kuiper Cliff, as well as account for a number of anomalies in the orbits and tilts of the ice giants.

The High Definition Space Telescope (HDST) is the next generation space telescope concept that will replace the James Webb Space Telescope (JWST). The Hubble Space Telescope has an aperture of 2.4 meters, while the JWST has an aperture of 6.5 meters. The HDST has a proposed aperture size of 11.7 meters, and scientists plan to deploy it at Sun-Earth L2 Lagrangian point. This is a telescope that can revolutionise studies into the Kuiper Belt, allowing for better studies of the population and densities of the KBOs by studying the region in unprecedented detail. At current capabilities, an object of about 10 km in diameter can be detected in the inner regions of the Kuiper Belt, and Dwarf Planets with a diameter of more than 1,000 kilometers can be observed at a distance of 500 AU, which is less than half of the distance at which Sedna, a minor planet in the outer reaches of the Solar System, spends most of its orbit. This telescope will have the capability to study the Kuiper Belt in a higher resolution, allowing for a detection of KBOs that are as small as 1-1.5 km in diameter. It can conduct spectroscopic analysis to investigate the composition of objects up to 10 km in diameter within the Kuiper Belt, and can detect any objects 500 km in diameter and larger throughout the outer regions of the scattered disk that extends to a distance of 1,000 AU. The HDST will have the capability to detect Earth to Neptune sized planets up to a distance of 15000 AU, and finally put to rest any claims of another large planet in the Solar System, beyond the orbit of Pluto. The HDST is expected to be deployed in the mid 2030s, and will be able to solve the mystery of the Kuiper Cliff. 